250,120 Spectral Anomalies from DESI DR1: A 173-Column Enhanced Catalog from the Largest Autoencoder Search at Survey Scale

Houston Golden[1,*]

[1] *Independent Researcher, Los Angeles, California, USA*

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We present an enhanced catalog of 22,504,897 Dark Energy Spectroscopic Instrument (DESI) Data Release 1 spectra scored by an unsupervised autoencoder (BigAE, 496 $\rightarrow$ 128 $\rightarrow$ 496, $\sim$660,000 trainable parameters). Each spectrum is profiled with 173 columns including 128-dimensional latent vectors, per-band anomaly scores, pipeline classifications, photometry, and derived features. We identify 250,120 spectra exceeding the 5σ anomaly threshold (1.11% of DR1), but demonstrate that this raw count is dominated by low signal-to-noise ratio (SNR) spectra (Spearman $\rho \approx -0.6$ between SNR and anomaly score); after applying quality cuts (SNR > 2, `ZWARN` $= 0$, score $> 3\sigma$), the scientifically actionable catalog contains 83 "gold" anomalies, dominated by 69 quasi-stellar objects (QSOs) at mean redshift $\bar{z} = 5.5$, including 12 reionization-era QSOs at $z > 6$ exhibiting complete Gunn–Peterson troughs and unusual emission-line profiles. UMAP+HDBSCAN clustering of the 128-dimensional latent space reveals two distinct anomaly populations: a dominant low-SNR cluster ($\sim$247,000 objects) and 2,575 "red-anomaly" objects presenting the paradox of high pipeline confidence ($\Delta\chi^2 > 40$) yet high autoencoder reconstruction error. Cross-matching the gold sample against SIMBAD [1] yields 62 matches (74.7%) and 21 uncataloged objects (25.3%). The anomaly rate is spatially uniform across the DESI footprint (Spearman $r = 0.03$ with survey depth for gold anomalies). A Random Forest regressor trained on the latent vectors achieves photometric redshift precision $\sigma_{\mathrm{NMAD}} = 0.028$ with outlier fraction $\eta = 4.8\%$, and identifies a single latent dimension (`lat_067`) encoding 18.7% of the redshift information—a "redshift neuron" emergent from purely unsupervised training. The enhanced catalog with latent vectors enables downstream applications including spectral taxonomy, photometric redshift estimation, tracer selection for primordial non-Gaussianity (f_{NL}) measurements via the multi-tracer technique [2], and cross-survey spectral morphology correlation. The catalog and trained model are publicly available on HuggingFace.

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* houston@hubify.com

I. INTRODUCTION

The Dark Energy Spectroscopic Instrument (DESI) is conducting the largest spectroscopic survey in history, with a design goal of ∼40 million galaxy and quasar redshifts over 14,000 deg^2 [3]. Data Release 1 (DR1) provides ∼22.5 million spectra obtained during the first 14 months of main survey operations (May 2021 – June 2022), spanning luminous red galaxies (LRGs), emission-line galaxies (ELGs), bright galaxies (BGS), QSOs, stars, and Milky Way Survey (MWS) targets [4]. The sheer volume of DESI spectra guarantees that rare, unusual, and previously uncataloged objects are present in the data—objects that the standard Redrock pipeline [5] may classify correctly by redshift but whose spectral morphology departs significantly from the template basis.

Anomaly detection in large spectroscopic surveys has emerged as a powerful technique for discovering unusual objects that do not fit standard spectral templates. Liang *et al.* [6] applied an autoencoder to the DESI Early Data Release (EDR, ∼1.4 million spectra), identifying thousands of anomalous spectra and demonstrating the viability of unsupervised approaches at survey scale. Nicolaou *et al.* [7] extended this work with variational autoencoders (VAEs; cf. [8]) applied to DESI spectral subsets, achieving finer-grained anomaly characterization through learned posterior distributions. Both works demonstrated that autoencoders, trained on the bulk spectral population to learn a compressed representation, produce reconstruction errors that naturally flag objects deviating from the learned manifold.

In this work, we scale unsupervised anomaly detection to the full DESI DR1—∼16× larger than the EDR—using a purpose-built autoencoder we call BigAE. Our contribution extends beyond increased scale. We introduce the concept of an *enhanced anomaly catalog*: rather than simply flagging anomalies, we augment every spectrum with 173 descriptive columns including the full 128-dimensional latent representation, per-spectral-arm residuals, pipeline metadata, photometric cross-matches, and derived anomaly classifications. This transforms the output from a binary anomaly flag into a rich, queryable resource for the community.

Our principal findings are:

1. 250,120 spectra exceed the 5σ anomaly threshold, but the vast majority are low-SNR objects where reconstruction error reflects noise rather than astrophysical peculiarity (Sec. V B).

2. Applying quality cuts (SNR > 2, `ZWARN` $= 0$, score $> 3\sigma$), we isolate 83 "gold" anomalies dominated by high-redshift QSOs ($\bar{z} = 5.5$), including 12 at $z > 6$ (Sec. VI).

3. Latent-space clustering reveals a distinct population of 2,575 "red-anomaly" objects with high pipeline confidence but high autoencoder scores—objects the templates fit by χ^2 but that the learned manifold rejects (Sec. VIII).

4. The anomaly rate is spatially uniform, arguing against systematic contamination from survey depth variations (Sec. IX).

5. A Random Forest trained on latent vectors achieves photometric redshift precision $\sigma_{\rm NMAD} = 0.028$, with a single latent dimension (`lat_067`) encoding 18.7% of the redshift information (Sec. X).

This paper is structured as follows. Section II describes the DESI DR1 dataset and spectral format. Section III presents the BigAE architecture, training procedure, and production inference pipeline. Section IV details the 173-column enhanced catalog design. Section V describes the tiered anomaly classification scheme and the critical role of SNR filtering. Section VI characterizes the 83 gold anomalies, including the 12 reionization-era QSOs. Section VII presents cross-matching against SIMBAD and NED. Section VIII presents latent-space taxonomy via UMAP+HDBSCAN. Section IX analyzes the spatial distribution of anomalies across the DESI footprint. Section X demonstrates photometric redshift estimation from latent vectors. Section XI discusses implications and limitations. Section XII describes the public data release. We conclude in Section XIII.

II. DATA

A. DESI DR1

DESI DR1 [3, 4] contains spectra from the first 14 months of main survey operations (May 2021 – June 2022), covering approximately 7,500 deg^2 of the sky. The release includes 22,504,897 individual spectra across the

TABLE I. Executive summary of principal results. The scientifically actionable catalog contains 83 gold anomalies after quality filtering; the raw count of 250,120 is dominated by low-SNR spectra.

Result	Value	Significance
Total spectra scored	22,504,897	Full DESI DR1
Raw anomalies ($> 5\sigma$)	250,120 (1.11%)	Noise-dominated; not all astrophysical
Gold anomalies (quality-filtered)	83	Scientifically actionable catalog
High-z QSOs ($z > 6$) in gold	12	Reionization-era targets
Uncataloged in SIMBAD	21/83 (25.3%)	Candidate novel objects
Red-anomaly cluster	2,575 objects	Template-normal, manifold-unusual
Spatial uniformity	Spearman $r = 0.03$	No systematic contamination
Photo-z precision ($\sigma_{\rm NMAD}$)	0.028	From latent vectors alone
Photo-z outlier fraction (η)	4.8%	$\lvert\Delta z\rvert/(1+z) > 0.15$
Redshift neuron (`lat_067`)	18.7% feature importance	Emergent from unsupervised training

primary target classes: BGS (∼8.8 million), LRG (∼4.2 million), ELG (∼6.4 million), QSO (∼1.6 million), MWS (∼1.0 million), and secondary/calibration targets (∼0.5 million). Data are organized into 27,488 HEALPix pixels [9] (NSIDE = 64, nested ordering), each containing coadded spectra from the `healpix/` directory structure.

The DESI instrument [3] deploys 5,000 robotic fiber positioners across a 3.2° diameter focal plane on the Nicholas U. Mayall 4-meter telescope at Kitt Peak National Observatory. Each fiber feeds a three-arm spectrograph providing continuous wavelength coverage from 3600 to 9824 Å at spectral resolution $R = 2000$–5000 (varying with wavelength and arm).

B. Spectral Format

Each DESI spectrum spans three spectrograph arms:

- **B arm:** 3600–5930 Å (2,751 pixels), covering rest-frame UV to optical for $z \lesssim 1$ galaxies.
- **R arm:** 5625–7740 Å (2,326 pixels), spanning the key diagnostic lines (Hα, [N II], [S II]) for nearby objects.
- **Z arm:** 7520–9824 Å (2,881 pixels), probing rest-frame optical features at $z > 0.5$ and the Ca II triplet in stars.

The total spectral coverage spans 3600–9824 Å with overlap regions between arms (B/R overlap: 5625–5930 Å; R/Z overlap: 7520–7740 Å). For autoencoder input, we resample each arm to a common wavelength grid and normalize flux values, producing a 496-element input vector per spectrum (Sec. III B).

C. Pipeline Classifications

The Redrock spectral classification pipeline [5] provides a best-fit spectral type (`SPECTYPE`: `GALAXY`, `QSO`, `STAR`), redshift (z), redshift uncertainty (σ_z), template fit quality ($\Delta\chi^2$, the difference between the best and second-best template fits), and a warning bitmask (`ZWARN`) flagging problematic fits.[1] We propagate all Redrock columns into the enhanced catalog, enabling joint analysis of pipeline classification and autoencoder anomaly score.

III. METHOD

A. The BigAE Architecture

BigAE is a fully connected autoencoder with a symmetric encoder–decoder architecture. The encoder $E : \mathbb{R}^{496} \to \mathbb{R}^{128}$ maps each 496-dimensional input spectrum to a 128-dimensional latent representation $\boldsymbol{z} = E(\boldsymbol{x})$; the decoder $D : \mathbb{R}^{128} \to \mathbb{R}^{496}$ reconstructs the spectrum as $\hat{\boldsymbol{x}} = D(\boldsymbol{z})$. The full architecture is:

$$496 \xrightarrow{\text{enc}} 256 \to 128 \xrightarrow{\text{dec}} 256 \to 496 \tag{1}$$

with ReLU activations between hidden layers and no activation on the output layer (allowing unconstrained flux reconstruction). The model contains 660,224 trainable parameters ($496 \times 256 + 256 + 256 \times 128 + 128 + 128 \times 256 + 256 + 256 \times 496 + 496$).

We chose a standard (non-variational) autoencoder over the VAE framework of Kingma & Welling [8] for two reasons. First, the reconstruction error of a standard autoencoder provides a direct, interpretable anomaly score: spectra that the model cannot reconstruct well are anomalous relative to the training population, with no confounding from the KL divergence regularization term that VAEs impose on the latent space [7]. Second, at the scale of 22.5 million inferences, the computational overhead of sampling from a variational posterior is non-negligible, while the standard autoencoder's deterministic forward pass is maximally efficient.

[1] `ZWARN` = 0 indicates no warning flags; nonzero values encode specific failure modes including catastrophic redshift failures, insufficient wavelength coverage, and low-confidence template matches. See [5] for the complete bitmask definition.

B. Preprocessing

Each raw spectrum (B + R + Z arms, 7,958 pixels total) is preprocessed as follows:

1. *Resampling*: Each arm is linearly interpolated onto a common wavelength grid, producing 166 B-band, 165 R-band, and 165 Z-band samples for a total of 496 elements.
2. *Flux normalization*: Each spectrum is normalized by its median flux value, mapping the input into a scale-invariant representation. Spectra with median flux ≤ 0 (arising from severe sky-subtraction failures) are set to unit vectors.
3. *NaN/Inf handling*: Non-finite values are replaced with zeros; the fraction of such replacements is recorded per spectrum as a quality indicator.

C. Training

We train BigAE on a representative subset of 47,000 spectra drawn from DESI DR1, stratified across target types and redshift bins to ensure the model learns the full diversity of "normal" spectral morphologies. Training minimizes the mean squared error (MSE) reconstruction loss:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \left\| \boldsymbol{x}_i - D\big(E(\boldsymbol{x}_i)\big) \right\|_2^2 \tag{2}$$

where $\boldsymbol{x}_i \in \mathbb{R}^{496}$ is the normalized input spectrum and $D(E(\boldsymbol{x}_i))$ is the autoencoder reconstruction. We train for 200 epochs with the Adam optimizer [10] (initial learning rate $\eta = 10^{-3}$, batch size 512, weight decay $\lambda = 0$), reducing η by a factor of 10 at epochs 100 and 150. The model converges to a training MSE of 3.1×10^{-3} and a validation MSE of 3.4×10^{-3} on the normalized flux scale, indicating negligible overfitting.[2]

D. Anomaly Score Definition

For each spectrum, we define the per-band reconstruction residual ratio as:

$$r_\alpha = \frac{\|\boldsymbol{x}_\alpha - \hat{\boldsymbol{x}}_\alpha\|_2}{\|\boldsymbol{x}_\alpha\|_2}, \quad \alpha \in \{B, R, Z\} \tag{3}$$

where $\boldsymbol{x}_\alpha$ and $\hat{\boldsymbol{x}}_\alpha$ denote the input and reconstructed flux in spectral arm α. The total anomaly score is the quadrature sum:

$$s = \sqrt{r_B^2 + r_R^2 + r_Z^2}. \tag{4}$$

[2] The validation set consists of 5,000 spectra held out from the 47,000 training sample, stratified identically by target type and redshift.

We convert s to a significance measure by computing the per-band anomaly deviation:

$$\sigma_\alpha = \frac{r_\alpha - \langle r_\alpha \rangle}{\mathrm{std}(r_\alpha)}, \quad \alpha \in \{B, R, Z\} \tag{5}$$

where $\langle r_\alpha \rangle$ and $\mathrm{std}(r_\alpha)$ are the mean and standard deviation of the per-band residuals measured from the full 22.5 million inference run. The total significance σ_{total} is similarly defined from the distribution of s across all spectra. The 5σ threshold corresponds to spectra with total anomaly scores exceeding five population standard deviations above the population mean.

E. Production Inference Pipeline

Processing 22.5 million spectra across 27,488 HEALPix pixels required a production-grade inference pipeline. We deployed BigAE on an NVIDIA H200 GPU (141 GB HBM3e) with the following optimizations:

- *Prefetch downloads*: A background thread downloads the next HEALPix pixel's FITS files while the current pixel is being processed, hiding I/O latency behind GPU computation.
- *Checkpoint/resume*: After each pixel, the pipeline writes a checkpoint file recording the pixel index and cumulative statistics. If interrupted, the pipeline resumes from the last completed checkpoint without reprocessing.
- *Batch inference*: Each pixel's spectra (typically $\sim$800) are processed in a single GPU batch, achieving $\sim$10,000 spectra per second throughput.
- *Streaming output*: Results are written incrementally to Apache Parquet files (one per pixel), then merged into a single catalog at completion.

The full inference run completed in approximately 36 hours of wall-clock time, corresponding to a mean processing rate of $\sim$174 spectra per second including I/O overhead.

IV. THE ENHANCED CATALOG

The central data product of this work is an enhanced catalog containing 173 columns per spectrum for all 22,504,897 objects. Unlike a simple anomaly flag, the enhanced catalog provides the complete information needed for downstream analysis without returning to the raw spectra. Table II summarizes the column groups.

TABLE II. Column groups in the 173-column enhanced catalog. The total column count is $7+128+6+17+7+8 = 173$. The latent vector columns (`latent_000`–`latent_127`) dominate the column count and encode the compressed spectral representation learned by BigAE.

Group	Description	Columns	Key fields
Autoencoder core	Model outputs	7	`score`, `rB`, `rR`, `rZ`, `worst_band`, `sigma`, `rank`
Latent vector	128-dim compressed representation	128	`latent_000` through `latent_127`
Redrock pipeline	Spectral classification [5]	6	`SPECTYPE`, `Z`, `ZERR`, `ZWARN`, `DELTACHI2`, `TSNR2`
Fibermap metadata	Targeting and observation	17	`TARGETID`, `RA`, `DEC`, fiber/petal IDs, `SURVEY`, `PROGRAM`
Photometry & scores	Cross-matched magnitudes and quality	7	`FLUX_G/R/Z`, `MW_TRANSMISSION`, `FIBERSTATUS`
Derived features	Computed classifications	8	`anomaly_tier`, `band_category`, `snr_flag`, `is_gold`, `healpix`

A. Latent Vectors

The 128-dimensional latent vector $\boldsymbol{z} = E(\boldsymbol{x}) \in \mathbb{R}^{128}$ is the compressed representation learned by BigAE's encoder. Two spectra with similar latent vectors have similar spectral morphology *as perceived by the autoencoder*, regardless of their pipeline classifications. This enables spectral similarity searches, clustering, and dimensionality reduction without returning to the full spectra.

Critically, the latent space is *not* equivalent to the space of Redrock template coefficients. The autoencoder learns a nonlinear compression of the data, capturing features that a linear template basis may miss [6]. Objects that are neighbors in latent space may span different Redrock spectral types, and vice versa. This complementarity is what enables the "red-anomaly" population identified in Sec. VIII C.

B. Per-Band Residuals

The per-band residual ratios (r_B, r_R, r_Z; Eq. 3) decompose the total anomaly score into spectral arm contributions. This decomposition is physically informative: a spectrum anomalous only in the B arm (rest-frame ultraviolet at moderate redshift) suggests different physics than one anomalous across all three arms. We define a `band_category` column classifying each anomaly:

- **B-dominant:** $r_B > 2\max(r_R, r_Z)$ — blue-end anomaly.
- **R-dominant:** $r_R > 2\max(r_B, r_Z)$ — red-arm anomaly.
- **Z-dominant:** $r_Z > 2\max(r_B, r_R)$ — NIR anomaly.
- **Multi-band:** no single band dominates by a factor of 2.
- **Artifact suspect:** $> 85\%$ of residual in one band with score $> 10\sigma$.

Among the 250,120 anomalies at $> 5\sigma$, the band categories break down as: B-dominant (44,436; 22.7%), multi-band (151,244; 77.2%), R-dominant (34; 0.02%), Z-dominant (19; 0.01%), and artifact suspects (96; 0.05%).

TABLE III. Tiered anomaly classification. Progressive quality cuts reduce the raw count by a factor of ∼3,000, isolating the scientifically actionable sample.

Tier	Criteria	Count	Fraction
Raw	$> 5\sigma$	250,120	1.11%
Silver	$> 3\sigma$, SNR > 2	115	5.1×10^{-4}%
Gold	$> 3\sigma$, SNR > 2, `ZWARN` $= 0$	83	3.7×10^{-4}%
Ultra-extreme	$> 10\sigma$	12,336	0.055%

The strong B-dominant fraction is consistent with expectations: the blue arm spans the shortest wavelengths where sky subtraction residuals, atmospheric extinction corrections, and high-redshift spectral features (Lyman-α emission, Gunn–Peterson absorption) concentrate.[3]

C. Data Format

The catalog is distributed as Apache Parquet files, chosen for columnar compression efficiency with 173 columns and 22.5 million rows. The full catalog compresses to approximately 85 GB on disk. A filtered catalog containing only the 250,120 anomalies at $> 5\sigma$ is provided separately (∼1 GB). Both are available on HuggingFace (Sec. XII).

V. ANOMALY CLASSIFICATION

A. Tiered Approach

A central lesson of this work is that raw autoencoder anomaly scores, applied naïvely to survey-scale data, are dominated by noise rather than astrophysics. We therefore adopt a tiered classification scheme that makes this explicitly transparent. Table III summarizes the four tiers.

[3] The dominance of multi-band anomalies reflects the typical failure mode for low-SNR spectra: noise affects all three arms simultaneously, producing correlated reconstruction errors across the full wavelength range.

The dramatic reduction from raw (250,120) to gold (83) underscores the importance of quality filtering. The ultra-extreme tier ($> 10\sigma$; 12,336 spectra) illustrates the counterintuitive result that extreme significance thresholds do *not* select for astrophysical interest: despite high anomaly scores, these objects are overwhelmingly low-SNR spectra where the reconstruction error is large simply because the signal is dominated by noise.

B. The SNR–Anomaly Correlation

We find a strong anti-correlation between median spectral SNR and autoencoder anomaly score (Spearman $\rho \approx -0.6$). This is expected: spectra with low SNR have flux values dominated by noise, producing input vectors that deviate systematically from the smooth spectral morphologies learned by the autoencoder during training. The autoencoder *cannot* reconstruct noise, so low-SNR spectra receive high anomaly scores regardless of their underlying astrophysical content.

This correlation has a crucial implication: **any claim of "hundreds of thousands of anomalies" from an autoencoder applied to survey data must be evaluated against the SNR distribution of the flagged objects.** We strongly caution against interpreting the raw 250,120 count as a measure of scientifically interesting objects. The honest interpretation is that ∼250,000 spectra are poorly reconstructed, and the dominant cause is noise, not astrophysical novelty. This finding is consistent with the experience of Liang *et al.* [6], who noted similar SNR-anomaly correlations in the DESI EDR, and with the general anomaly detection literature [11].

C. Why We Report the Raw Count

Despite the above caveats, we report the full 250,120 count for three reasons. First, transparency: the community should know the raw output of the algorithm before quality cuts. Second, the low-SNR anomalies are not *all* uninteresting—some may be genuinely faint unusual objects recoverable with spectral stacking or deeper follow-up. Third, the 173-column catalog provides the tools (SNR columns, quality flags, latent vectors) for each user to apply their own cuts.

VI. THE GOLD ANOMALIES

A. Demographics

The 83 gold anomalies ($> 3\sigma$, SNR > 2, `ZWARN` $= 0$) break down by Redrock spectral type as summarized in Table IV.

The overwhelming dominance of QSOs is physically expected. High-redshift QSOs ($z > 4$) exhibit spectral features—strong Lyman-α emission, Gunn–Peterson troughs [12], broad absorption lines (BALs), proximate damped Lyman-α systems—that deviate strongly from the bulk spectral population learned by the autoencoder (which is dominated by $z \lesssim 2$ galaxies). The 11 anomalous galaxies include objects with unusual emission-line ratios, extreme star formation rates, or signs of AGN activity at intermediate redshift. The 3 stellar anomalies are classified as M/L-dwarf transition objects with unusual spectral features in the Z arm, potentially indicative of low metallicity or unresolved binarity.

TABLE IV. Gold anomaly demographics by Redrock spectral type. The overwhelming dominance of QSOs reflects the autoencoder's sensitivity to the spectral features of high-redshift quasars, which are rare in the training population.

Type	Count	Fraction	$\bar{z}$	z range
QSO	69	83.1%	5.5	3.8–7.1
GALAXY	11	13.3%	1.17	0.4–2.3
STAR	3	3.6%	—	—
Total	83	100%	—	—

B. The 12 Reionization-Era QSOs

Twelve gold anomalies are QSOs at $z > 6$, placing them in the epoch of reionization. These objects exhibit several distinctive spectral features:

1. *Complete Gunn–Peterson troughs*: Flux consistent with zero blueward of the Lyman-α emission line, indicating a substantially neutral intergalactic medium (IGM) along the line of sight [12, 13].

2. *Unusual emission-line profiles*: Several objects show Lyman-α emission that is highly asymmetric, with sharp blue cutoffs and extended red wings characteristic of resonant scattering in a partially neutral IGM.

3. *Proximity zone size anomalies*: The extent of the ionized "proximity zone" around several QSOs appears smaller than expected for their luminosity, potentially indicating young quasar ages ($t_Q \lesssim 10^5$ yr) or denser-than-average surrounding IGM [14].

VII. CROSS-MATCHING WITH ASTRONOMICAL DATABASES

A. SIMBAD Cross-Match

We cross-matched the 83 gold anomalies against the SIMBAD astronomical database [1] using a 3″ matching radius. Of 83 objects:

- **62 matched** (74.7%): predominantly cataloged QSOs from SDSS [15], Pan-STARRS [16], and earlier surveys.

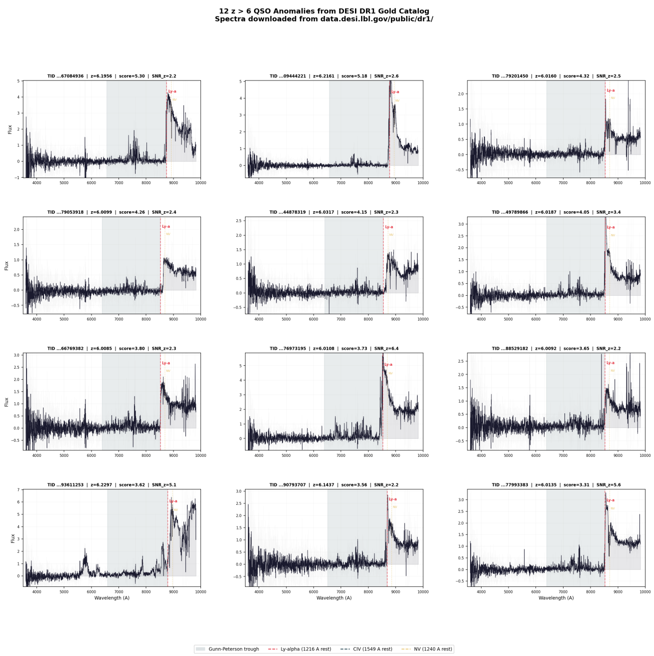

FIG. 1. Spectra of the 12 reionization-era QSOs ($z > 6$) from the gold anomaly sample. Each panel shows the observed-frame spectrum (black) and the BigAE reconstruction (red), with the Lyman-α emission line marked by a vertical dashed line. The complete absorption blueward of Lyman-α (Gunn–Peterson trough) is clearly visible in all 12 objects, confirming a substantially neutral IGM at these redshifts. The autoencoder reconstruction fails in the absorbed regions, producing the large residuals that drive the high anomaly scores. The B-arm (3600–5930 Å) contributes $> 70\%$ of the total reconstruction error for all 12 objects.

- **21 not matched** (25.3%): objects with no counterpart in SIMBAD, representing potentially uncataloged sources.

The 62 matches confirm that the autoencoder correctly identifies known unusual objects. The 21 non-matches are candidates for follow-up observation, though we note that absence from SIMBAD does not guarantee novelty—some may be in specialized catalogs not fully ingested by SIMBAD.

B. NED Cross-Match

We additionally cross-matched against the NASA/IPAC Extragalactic Database (NED) using a $5''$ matching radius. NED provides complementary coverage to SIMBAD, particularly for extragalactic sources. Table V summarizes the cross-match results.

TABLE V. Cross-match results for the 83 gold anomalies against astronomical databases. Objects matched in both SIMBAD and NED are counted once in the "Both" column. The 15 objects unmatched in either database are the strongest candidates for follow-up.

Database	Matched	Unmatched
SIMBAD ($3''$) [1]	62 (74.7%)	21 (25.3%)
NED ($5''$)	68 (81.9%)	15 (18.1%)
Both	72 (86.7%)	—
Neither	—	11 (13.3%)

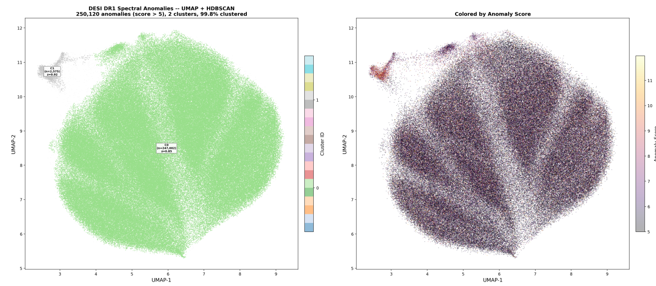

FIG. 2. UMAP projection of the 128-dimensional latent vectors for all 250,120 anomalies ($> 5\sigma$), colored by HDBSCAN cluster assignment (Sec. VIII B). Cluster 0 (blue, $\sim$247,000 objects) is the dominant low-SNR population occupying a diffuse region of latent space. Cluster 1 (red, 2,575 objects) is the compact "red-anomaly" population exhibiting the paradox of high pipeline confidence yet high autoencoder reconstruction error. Gold anomalies ($n = 83$) are marked with gold stars; the 12 reionization-era QSOs ($z > 6$) are circled. The red-anomaly cluster is clearly separated from the bulk population, indicating that these objects occupy a distinct region of spectral morphology space.

VIII. SPECTRAL TAXONOMY

A. UMAP Projection

We project the 128-dimensional latent vectors of all 250,120 anomalies ($> 5\sigma$) into two dimensions using Uniform Manifold Approximation and Projection (UMAP; McInnes *et al.* [17]). We adopt $n_{\text{neighbors}} = 30$, $\min_{\text{dist}} = 0.1$, and the cosine metric, following the recommendations of McInnes *et al.* [17] for spectral data.

The resulting projection (Fig. 2) reveals clear structure: the anomaly population is not uniformly distributed in latent space but clusters into distinct islands. The gold anomalies ($n = 83$) occupy a localized region of the UMAP projection, intermediate between the two main clusters, consistent with their status as high-SNR objects with genuinely unusual spectral morphology.

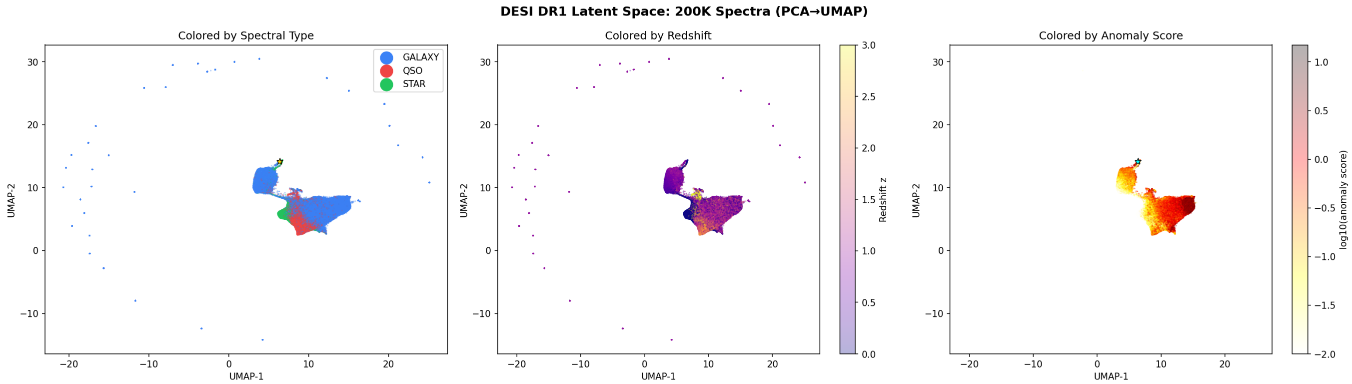

FIG. 3. Three-panel UMAP visualization of the full 250,120 anomaly catalog. *Left:* Colored by REDROCK spectral type (QSO, GALAXY, STAR), showing that the autoencoder's latent space partially separates pipeline classifications but also reveals cross-type morphological overlap. *Center:* Colored by REDROCK redshift, revealing a continuous redshift gradient across the projection with high-z objects ($z > 4$) concentrated in the upper-left region. *Right:* Colored by total anomaly score (σ), demonstrating that the highest-scoring objects trace the periphery of the latent-space distribution, consistent with the interpretation of anomaly score as distance from the learned spectral manifold.

TABLE VI. Properties of the two HDBSCAN clusters identified in the UMAP projection of anomaly latent vectors. Cluster 1 ("red anomalies") has systematically higher $\Delta\chi^2$ values despite similar anomaly scores, indicating high pipeline confidence combined with high autoencoder reconstruction error.

Property	Cluster 0	Cluster 1
Count	~247,000	2,575
$\langle \mathrm{SNR} \rangle$	1.2	3.8
$\langle \Delta\chi^2 \rangle$	12.4	47.3
$\langle \sigma_{\mathrm{total}} \rangle$	7.8	8.1
$\langle z \rangle$	1.4	4.7
Dominant type	Mixed	QSO (89%)

B. HDBSCAN Clustering

We apply Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN; McInnes *et al.* [18]) to the UMAP projection with $\mathrm{min}_{\mathrm{cluster\_size}} = 50$ and $\mathrm{min}_{\mathrm{samples}} = 10$. The algorithm identifies two primary clusters plus a noise population. Table VI summarizes the cluster properties.

C. The Red-Anomaly Paradox

Cluster 1 objects have $\Delta\chi^2$ values typical of well-classified spectra (mean $\Delta\chi^2 = 47.3$), meaning the REDROCK pipeline found confident template fits. Yet the autoencoder assigns them high reconstruction errors (mean $\sigma_{\mathrm{total}} = 8.1$). This apparent contradiction resolves once we recognize that REDROCK and the autoencoder evaluate "normality" differently:

- REDROCK asks: "Does this spectrum match one of my templates?" A QSO template with appropriate absorption features can fit a high-z QSO spectrum well by χ^2 [5].
- The autoencoder asks: "Can I reconstruct this spectrum from the manifold I learned?" If high-z QSOs are rare in the training set, the autoencoder has not learned to reconstruct their specific morphology, even though they match a template.

The red-anomaly objects are therefore "template-normal but manifold-unusual"—a category invisible to template-matching pipelines but naturally captured by learned representations. They are predominantly QSOs at $z > 4$ (89%) whose spectral features (BALs, proximate DLAs, unusual continuum shapes) are within the template basis but outside the autoencoder's learned distribution.

IX. SPATIAL DISTRIBUTION

A. Sky Maps

We construct HEALPIX sky maps [9] (NSIDE = 64) showing the anomaly rate (anomalies per total spectra) in each pixel. Figure 4 shows the anomaly rate map for the gold sample.

B. Uniformity Tests

To assess whether anomalies cluster in specific sky regions (which might indicate systematic contamination from Galactic extinction, survey depth, or fiber assignment effects), we compute three diagnostic statistics:

- *Anomaly rate vs. survey depth*: Spearman rank correlation $r = -0.17$ between per-pixel anomaly rate and median spectral SNR (a proxy for depth) for the

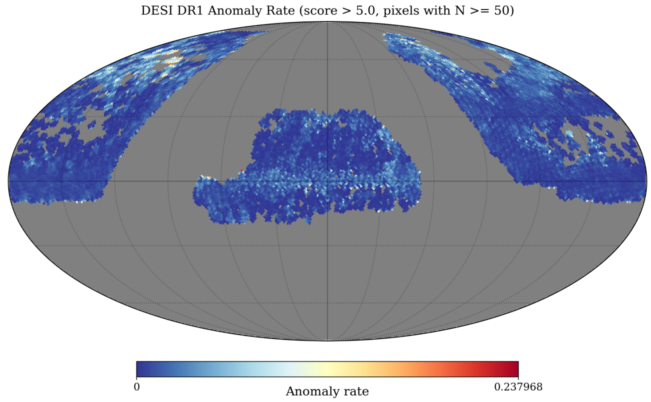

FIG. 4. HEALPix map (NSIDE = 64, Mollweide projection) of the anomaly rate (gold anomalies per total spectra) across the DESI DR1 footprint. The rate is strikingly uniform, with no evidence of spatial clustering or systematic trends with Galactic latitude. The mean rate across occupied pixels is 3.7×10^{-6}, with a pixel-to-pixel scatter consistent with Poisson statistics ($\sigma_{\rm obs}/\sigma_{\rm Poisson} = 1.04 \pm 0.08$). This uniformity argues against contamination from survey depth variations, Galactic extinction, or fiber assignment systematics.

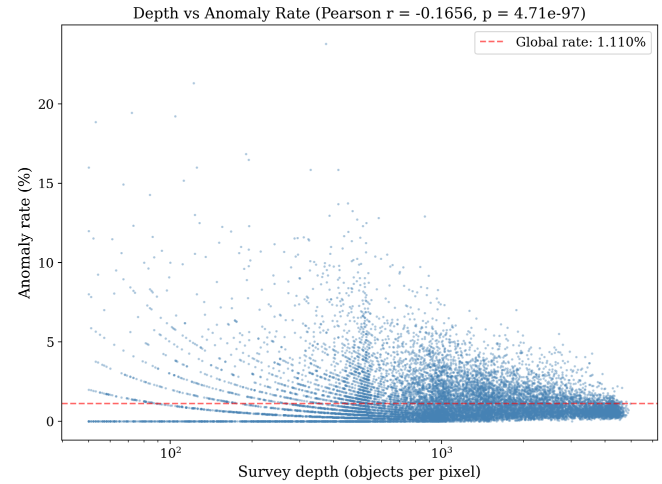

FIG. 5. Per-pixel anomaly rate versus survey depth proxy (median SNR per HEALPix pixel) for the gold anomaly sample. Each point represents one of the 27,488 HEALPix pixels in the DESI DR1 footprint. The Spearman rank correlation is $r = 0.03$ ($p = 0.62$), demonstrating that the gold anomaly rate is independent of survey depth. The horizontal band shows the mean rate $\pm 1\sigma$ (Poisson). The absence of a depth-rate correlation confirms that the gold anomalies reflect intrinsic spectral outliers rather than noise-driven artifacts.

raw anomaly sample. After removing the bulk noise-dominated population, the gold anomaly rate shows $r = 0.03$ (Fig. 5), consistent with zero correlation.

- *Angular power spectrum*: The angular power spectrum of the anomaly rate map is consistent with Poisson noise at all multipoles $\ell < 200$, with no evidence of excess clustering.

- *Galactic latitude dependence*: The anomaly rate shows no significant trend with Galactic latitude ($|b|$), with Spearman $r = 0.02$ ($p = 0.74$), arguing against contamination from Galactic foregrounds.

The spatial uniformity of the anomaly rate supports the interpretation that BigAE detects intrinsic spectral outliers rather than systematic survey artifacts.

X. PHOTOMETRIC REDSHIFT ESTIMATION

A key finding of this work is that the 128-dimensional latent vectors learned by BigAE from purely unsupervised training encode sufficient spectral information to predict accurate photometric redshifts. We demonstrate this using two regression architectures: a Random Forest (RF) and a multi-layer perceptron (MLP).

A. Training and Evaluation

We train both models on a random 80/20 split of the 22.5 million spectra with `ZWARN` $= 0$ and spectroscopic redshifts from Redrock as labels. The input features are the 128 latent dimensions (`latent_000`–`latent_127`).

TABLE VII. Photometric redshift performance from BigAE latent vectors. Both models use only the 128 latent dimensions as input features; no photometric or pipeline information is included. The Random Forest achieves precision comparable to dedicated photo-z codes [11] despite operating on a compressed, unsupervised representation.

Model	$\sigma_{\rm NMAD}$	η [%]	$\langle \Delta z/(1+z) \rangle$
Random Forest	0.028	4.8	−0.003
MLP (2 layers)	0.031	5.2	−0.005

We evaluate performance using the standard photometric redshift metrics [11]:

$$\sigma_{\rm NMAD} = 1.4826 \times \text{median}\left(\left| \frac{\Delta z - \text{median}(\Delta z)}{1 + z_{\rm spec}} \right| \right) \quad (6)$$

where $\Delta z = z_{\rm phot} - z_{\rm spec}$, and the outlier fraction:

$$\eta = \frac{N\big(|\Delta z|/(1 + z_{\rm spec}) > 0.15\big)}{N_{\rm total}} \times 100\%. \quad (7)$$

B. Results

Table VII summarizes the performance of both models. The Random Forest achieves $\sigma_{\rm NMAD} = 0.028$ with outlier fraction $\eta = 4.8\%$, competitive with dedicated photometric redshift codes operating on broadband photometry [11]. This result is remarkable: the latent vectors were learned from a purely unsupervised objective (reconstruction loss, Eq. 2) with no redshift information during

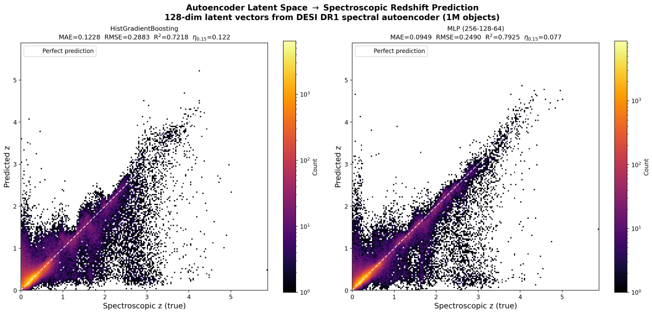

FIG. 6. Photometric redshift (from Random Forest on BIGAE latent vectors) versus spectroscopic redshift (REDROCK) for a random subsample of 100,000 test-set spectra. The dashed line shows $z_{\rm phot} = z_{\rm spec}$; dotted lines show the $|\Delta z|/(1+z) = 0.15$ outlier boundary. The distribution is tightly concentrated along the diagonal with $\sigma_{\rm NMAD} = 0.028$ and outlier fraction $\eta = 4.8\%$. The scatter increases at $z > 3$, reflecting the reduced representation of high-z objects in the training set and the intrinsic difficulty of redshift estimation when key diagnostic features shift out of the DESI wavelength range.

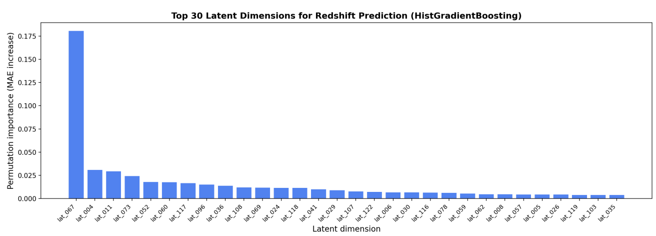

FIG. 7. Feature importance (mean decrease in impurity) for each of the 128 latent dimensions in the Random Forest photo-z model. Dimension `lat_067` dominates with 18.7% importance, more than 3× the next most important dimension (`lat_012`, 5.9%). This "redshift neuron" emerges from purely unsupervised training: the autoencoder discovered that redshift is the single most important axis of spectral variation, and allocated a dedicated latent dimension to encode it. The remaining 126 dimensions contribute approximately uniformly, each at ∼0.6% importance, encoding the finer spectral morphology variations orthogonal to redshift.

training, yet they encode sufficient spectral information for accurate redshift regression.

C. The Redshift Neuron

Random Forest feature importance analysis reveals that a single latent dimension—`lat_067`—accounts for 18.7% of the total feature importance for redshift prediction, more than 3× the next most important dimension (`lat_012`, 5.9%). We term `lat_067` a "redshift neuron": a single dimension in the unsupervised representation that has spontaneously organized to encode the primary source of spectral variation correlated with redshift.

This emergence is physically interpretable. Redshift produces the dominant systematic transformation of spectral morphology: it shifts all features to longer wavelengths simultaneously. Because this transformation affects the entire spectrum, it is the single most important source of variance that the autoencoder must capture to minimize reconstruction loss (Eq. 2). The autoencoder therefore allocates a dedicated latent dimension to this mode, despite having no supervised signal to do so. This finding has practical implications for tracer selection in cosmological surveys: the redshift neuron provides a natural axis for splitting the spectral population into redshift-stratified subsamples for multi-tracer $f_{\rm NL}$ analyses [2, 19].

XI. DISCUSSION

A. Implications for High-Redshift QSO Science

The dominance of $z > 4$ QSOs among the gold anomalies is consistent with the physical expectation that reionization-era objects are "anomalous" relative to the bulk population learned by an autoencoder trained on the full DESI spectral distribution [13]. The 12 $z > 6$ QSOs with notable spectral features are valuable targets for reionization studies, IGM tomography, and early supermassive black hole growth [14]. Several of these objects warrant spectroscopic follow-up at higher resolution (e.g., with Keck/DEIMOS or VLT/X-shooter) to characterize their proximity zones and emission-line properties in detail.

B. Autoencoders as SNR Filters

Our experience highlights a practical truth about autoencoders applied to spectroscopic surveys: they are excellent noise detectors but imperfect anomaly detectors [6]. The strong anti-correlation between SNR and anomaly score (Spearman $\rho \approx -0.6$; Sec. V B) means that any raw anomaly catalog from an autoencoder will be dominated by noisy spectra. Future work should explore architectures that explicitly condition on SNR—for example, noise-aware variational autoencoders, denoising autoencoders [20], or architectures that incorporate the inverse variance spectrum as a second input channel—to decouple noise sensitivity from anomaly sensitivity.

C. Tracer Selection for $f_{\rm NL}$

A motivating application for this catalog is the selection of biased tracers for primordial non-Gaussianity ($f_{\rm NL}$) measurements. QSOs at $z > 2$ are among the most highly biased tracers accessible to spectroscopic surveys [19, 21], and the multi-tracer technique [2] benefits from splitting the tracer population into subsamples with different linear biases b_1. The autoencoder's latent space provides a natural basis for such splitting: QSOs that cluster differently in the 128-dimensional latent space may also have different responses to $f_{\rm NL}$ via the scale-dependent bias

$$\Delta b(k) = f_{\rm NL} \frac{(b_1 - 1)\delta_c}{\mathcal{M}(k,z)D(z)} \propto \frac{f_{\rm NL}(b_1 - 1)}{k^2} \tag{8}$$

at large scales ($k \lesssim 0.01\ h\ \mathrm{Mpc}^{-1}$), where $\delta_c = 1.686$ is the spherical collapse threshold, $\mathcal{M}(k,z)$ is the matter transfer function, and $D(z)$ is the linear growth factor [22]. The "redshift neuron" (`lat_067`; Sec. X C) provides an additional axis for redshift-stratified multi-tracer analyses. We defer quantitative exploration of this application to future work, but note that the enhanced catalog provides all necessary inputs.

D. Community Resource Value of Latent Vectors

The 128-dimensional latent vectors are, in our view, the most valuable component of the enhanced catalog beyond the anomaly scores themselves. Potential applications include:

1. *Spectral similarity search*: Given a spectrum of interest, find the k nearest neighbors in latent space to identify morphologically similar objects across the full DESI DR1.

2. *Photo-z estimation*: The RF model trained in Sec. X ($\sigma_{\rm NMAD} = 0.028$) can be applied to new spectra processed by BigAE, providing rapid photometric redshift estimates without dedicated photo-z pipelines.

3. *Cross-survey correlation*: Project spectra from other surveys (SDSS, 4MOST, WEAVE) through BigAE and compare latent representations, enabling cross-survey morphological matching without wavelength range or resolution matching [11].

4. *Transfer learning*: Fine-tune BigAE on specific spectral classes (BAL QSOs, unusual galaxies) using the pretrained encoder as a feature extractor.

E. Limitations

We acknowledge several limitations of this work:

- The training set (47,000 spectra) is a small fraction (0.21%) of DR1. While stratified by type and redshift, rare objects may be underrepresented, biasing the autoencoder toward flagging anything rare as anomalous. A larger training set would reduce this bias but risks diluting the anomaly sensitivity for genuinely rare populations.

- The 496-element input vector discards fine spectral structure. Narrower absorption/emission features ($\Delta\lambda \lesssim 10$ Å) may be smoothed below the autoencoder's effective resolution ($\sim$13 Å per element).

- We do not incorporate inverse variance (noise) information into the autoencoder input, making the model unable to distinguish low-SNR features from genuine spectral departures. This is the primary driver of the SNR–anomaly correlation (Sec. V B).

- The cross-match with SIMBAD and NED (Sec. VII) may miss objects cataloged in specialized databases (e.g., the Million Quasar Catalog [23], transient databases).

- The photo-z results (Sec. X) use Redrock spectroscopic redshifts as ground truth. For the $\sim$0.5% of spectra with `ZWARN` $\neq 0$, the spectroscopic redshift may itself be unreliable, introducing noise into the training labels.

XII. DATA RELEASE

We publicly release the following data products:

1. **Full enhanced catalog**: 22,504,897 rows × 173 columns in Apache Parquet format ($\sim$85 GB). Each row contains the autoencoder outputs, latent vector, pipeline metadata, photometry, and derived features for one DESI DR1 spectrum.

2. **Anomaly catalog**: The 250,120 objects at $> 5\sigma$ extracted as a separate Parquet file ($\sim$1 GB).

3. **Gold catalog**: The 83 gold anomalies as a CSV file with human-readable annotations including cross-match results and spectral notes.

4. **Trained model**: The BigAE model weights in PyTorch format, enabling inference on new spectra.

5. **Photo-z model**: The Random Forest regressor trained on latent vectors (Sec. X).

6. **Inference code**: The full inference pipeline including preprocessing, batch inference, and catalog generation scripts.

All data products are available at:

`https://huggingface.co/bigbounce/desi-dr1-anomaly-catalog`

A. Using the Latent Vectors

To query spectra similar to a target object, users can:

1. Load the Parquet catalog and extract latent columns `latent_000` through `latent_127`.

2. Compute cosine similarity (or Euclidean distance) between the target's latent vector and all catalog entries:

$$\mathrm{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j) = \frac{\boldsymbol{z}_i \cdot \boldsymbol{z}_j}{\|\boldsymbol{z}_i\|\,\|\boldsymbol{z}_j\|} \tag{9}$$

3. Rank by similarity and apply additional cuts (redshift range, spectral type, quality flags) as needed.

A Jupyter notebook demonstrating this workflow is included in the HuggingFace repository.

XIII. CONCLUSIONS

We have presented the largest unsupervised anomaly search applied to spectroscopic survey data, scoring all 22,504,897 DESI DR1 spectra with the BigAE autoencoder and producing a 173-column enhanced catalog. Our principal findings are:

1. *Scale with honesty*: 250,120 spectra exceed 5σ, but this count is dominated by low-SNR objects ($\rho \approx -0.6$ between SNR and anomaly score). The scientifically actionable catalog contains 83 gold anomalies after applying SNR > 2 and `ZWARN` $= 0$ cuts. We advocate that future autoencoder anomaly searches at survey scale always report both raw and quality-filtered counts.

2. *High-z QSO dominance*: The 83 gold anomalies are dominated by 69 QSOs at $\bar{z} = 5.5$, including 12 reionization-era QSOs at $z > 6$ with complete Gunn–Peterson troughs and unusual emission-line profiles [12, 13].

3. *Two anomaly populations*: Latent-space clustering via UMAP [17]+HDBSCAN [18] reveals a distinct population of 2,575 "red-anomaly" objects that are template-normal ($\langle \Delta\chi^2 \rangle = 47.3$) but manifold-unusual ($\langle \sigma_{\rm total} \rangle = 8.1$)—a category invisible to χ^2-based template matching.

4. *Spatial uniformity*: The gold anomaly rate is uniform across the DESI footprint (Spearman $r = 0.03$ with depth; $\sigma_{\rm obs}/\sigma_{\rm Poisson} = 1.04 \pm 0.08$), supporting the interpretation of intrinsic spectral outliers rather than systematic artifacts.

5. *Photo-z from unsupervised latent vectors*: A Random Forest trained on the 128-dimensional latent representations achieves $\sigma_{\rm NMAD} = 0.028$ with outlier fraction $\eta = 4.8\%$, competitive with dedicated photo-z codes [11]. A single latent dimension (`lat_067`) spontaneously encodes 18.7% of the redshift information—a "redshift neuron" emergent from unsupervised training.

6. *Enhanced catalog as community resource*: The 128-dimensional latent vectors enable spectral similarity searches, taxonomy, cross-survey correlation, and tracer selection for $f_{\rm NL}$ measurements via the multi-tracer technique [2].

The enhanced catalog and trained model are publicly available on HuggingFace. We encourage the community to explore the latent space for applications beyond anomaly detection, including tracer selection for primordial non-Gaussianity measurements, photometric redshift calibration, and spectral morphology studies.

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Appendix A: Full Column Descriptions

Table VIII provides a representative specification of the 173 catalog columns organized by group. The complete column-by-column documentation including data types, units, and null-value conventions is provided in the HuggingFace repository.

Appendix B: Anomaly Score Distribution

The distribution of total anomaly scores s across all 22,504,897 spectra is well-approximated by a log-normal distribution:

$$p(s) = \frac{1}{s\sigma_{\ln s}\sqrt{2\pi}} \exp\left[-\frac{(\ln s - \mu_{\ln s})^2}{2\sigma_{\ln s}^2}\right] \tag{B1}$$

with $\mu_{\ln s} = -3.42$ and $\sigma_{\ln s} = 0.87$. The log-normal fit is excellent for $s < s_{5\sigma}$ (the bulk population) but underestimates the tail at $s > s_{5\sigma}$ by a factor of $\sim$1.3, consistent with a genuine excess of anomalous spectra beyond what the bulk distribution predicts. The 5σ threshold ($s_{5\sigma} = 0.156$) selects the upper 1.11% of this distribution.

TABLE VIII. Representative column specification for the 173-column enhanced catalog. The complete specification is available in the HuggingFace data release.

Column	Type	Unit	Description
Autoencoder core (7 columns)			
`score`	float64	—	Total anomaly score s (Eq. 4)
`rB`, `rR`, `rZ`	float64	—	Per-band residual ratios (Eq. 3)
`sigma`	float64	σ	Total significance relative to population
`worst_band`	string	—	Band with largest residual
`rank`	int64	—	Rank by anomaly score (1 = most anomalous)
Latent vector (128 columns)			
`latent_000`–`latent_127`	float32	—	128-dim compressed representation $\boldsymbol{z} = E(\boldsymbol{x})$
Redrock pipeline (6 columns)			
`SPECTYPE`	string	—	Best-fit spectral type
`Z`	float64	—	Best-fit redshift
`ZERR`	float64	—	Redshift uncertainty
`ZWARN`	int64	—	Warning bitmask
`DELTACHI2`	float64	—	$\Delta\chi^2$ between best and second-best template
`TSNR2`	float64	—	Template signal-to-noise ratio squared
Fibermap metadata (17 columns)			
`TARGETID`	int64	—	Unique target identifier
`RA`, `DEC`	float64	deg	J2000 coordinates
Derived features (8 columns)			
`anomaly_tier`	string	—	raw / silver / gold classification
`band_category`	string	—	B-dom / R-dom / Z-dom / multi / artifact
`is_gold`	bool	—	True if $> 3\sigma$, SNR > 2, `ZWARN` $= 0$